L9a: European Option Pricing with Black--Scholes--Merton

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Objectives for Today

L8b treated a quoted premium as an input. Today we compute that premium for a contract exercisable only at expiration.

Today's concepts

- Explain risk-neutral valuation: separate a no-arbitrage pricing measure from a real-world return forecast.
- Price European calls and puts: evaluate the Black–Scholes–Merton formulas with consistent units and conventions.
- Check the premium: use put–call parity, nonnegativity, strike monotonicity, and Monte Carlo standard errors.

Lecture and Example Stay on the Same Problem

Lecture: European exercise, risk-neutral valuation, the BSM call and put formulas, input conventions, put–call parity, and numerical checks.

Example: Price the same European call and put by BSM and by risk-neutral Monte Carlo. Report simulation uncertainty, verify parity, and sweep both premiums across strike.

Next: L9b changes the exercise right. The European price from today becomes the benchmark for measuring the value of American early-exercise flexibility.

European Exercise Means One Terminal Payoff

A European contract can be exercised only at expiration T. With strike $K > 0$, terminal share price $S_T \geq 0$, and $x^+ := \max(x, 0)$, the per-share payoffs are

$$\boxed{h_c(S_T; K) = (S_T - K)^+, \qquad h_p(S_T; K) = (K - S_T)^+.}$$

- Premiums: C_0 for the call and P_0 for the put, both in USD/share at time 0.
- Deliverable: multiply by shares/contract only when converting a per-share value to account dollars.
- Scope: there is no early-exercise decision to optimize. L9b adds that decision for American contracts.

Risk Neutral Is a Pricing Measure, Not a Forecast

Let g_y be the continuously compounded risk-free rate, δ a continuous dividend yield, and $\sigma > 0$ annualized volatility. Under the BSM pricing measure $\mathbb{Q}$,

$$\frac{dS_t}{S_t} = (g_y - \delta)\, dt + \sigma\, dW_t^{\mathbb{Q}}.$$

A European payoff $h(S_T)$ has model value

$$\boxed{V_0 = e^{-g_y T}\mathbb{E}^{\mathbb{Q}}[h(S_T)].}$$

- The drift $g_y - \delta$ enforces no-arbitrage pricing under the model.
- No estimate of the asset's historical or expected real-world return enters this valuation.

The Closed Form Comes with Assumptions

Classical BSM assumes:

- frictionless continuous trading and no arbitrage;
- constant risk-free rate g_y, dividend yield δ, and volatility σ over the contract's life;
- continuous lognormal underlying-price paths; and
- the ability to trade the underlying and financing asset as required by the replicating hedge.

Real markets have jumps, discrete trading, volatility smiles, funding spreads, transaction costs, and constraints. A formula can be evaluated exactly while its model remains approximate.

Black--Scholes--Merton Prices the Pair

Define the dimensionless quantities

$$d_1 = \frac{\ln(S_0/K) + (g_y - \delta + \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}, \qquad d_2 = d_1 - \sigma\sqrt{T}.$$

With Φ the standard-normal CDF, the European premiums are

$$\boxed{C_0 = S_0 e^{-\delta T}\Phi(d_1) - K e^{-g_y T}\Phi(d_2)}$$

$$\boxed{P_0 = K e^{-g_y T}\Phi(-d_2) - S_0 e^{-\delta T}\Phi(-d_1).}$$

Both values are in USD/share. Set $\delta = 0$ for a non-dividend-paying underlying.

Input Conventions Are Part of the Formula

Before evaluating BSM, reconcile every input:

- S_0 and K: the same currency per share;
- T: time to expiration in years, with an explicit day-count convention;
- g_y: a continuously compounded annual rate;
- σ: annualized log-return volatility; and
- δ: a continuous proportional dividend yield.

A mathematically correct formula with inconsistent conventions produces a precisely computed wrong answer.

Put--Call Parity Checks the Closed Forms

The European call and put satisfy

$$\boxed{C_0 - P_0 = S_0 e^{-\delta T} - K e^{-g_y T}.}$$

- Economic meaning: call minus put replicates the dividend-adjusted share minus a bond paying K at expiration.
- Implementation use: compute the call and put directly, then require the parity residual to match the implementation's precision.
- Numerical caution: do not obtain a tiny premium by subtracting nearly equal rounded quantities; cancellation can produce a negative value.

Monte Carlo Is a Second Route to the Same Value

Draw M independent endpoints under $\mathbb{Q}$, discount each payoff $Y_k = e^{-g_y T} h(S_T^{(k)})$, and estimate

$$\widehat{V}_M = \frac{1}{M} \sum_{k=1}^{M} Y_k, \qquad \widehat{\mathrm{SE}}(\widehat{V}_M) = \frac{s_Y}{\sqrt{M}}.$$

- Compare $\widehat{V}_M - V_0$ in estimated standard-error units.
- Report the random-seed policy, number of samples, and standard error.
- A nominal 95% interval is not guaranteed to contain the closed form on every run; interval membership is a diagnostic, not an identity.

Strike Moves Calls and Puts in Opposite Directions

Holding S_0, T, g_y, δ, and σ fixed:

- Calls are nonincreasing in K: a higher purchase price can only reduce the terminal call payoff state by state.
- Puts are nondecreasing in K: a higher sale price can only increase the terminal put payoff state by state.
- Both premiums remain nonnegative: they are prices of nonnegative terminal payoffs.

The example notebook verifies these properties across strikes from 20 to 100 USD/share.

Four Checks Before Trusting the Premium

1. $C_0 \geq 0$ and $P_0 \geq 0$.
2. Put–call parity holds within the formula's rounding tolerance.
3. Calls decrease and puts increase as strike rises with other inputs fixed.
4. Monte Carlo discrepancies are ordinary when measured in simulation standard-error units.

These checks establish internal consistency under BSM. They do not prove that constant volatility, continuous paths, or the other assumptions describe a particular market well.

Summary

L9a prices European options end to end: terminal payoff, risk-neutral valuation, BSM closed form, and numerical validation.

- Risk neutral is a pricing measure: no real-world return forecast enters the premium.
- BSM prices European exercise: inputs and compounding conventions are part of the model specification.
- A premium needs checks: use direct formulas, parity, monotonicity, and Monte Carlo uncertainty together.

Next, L9b adds American exercise and asks when that extra right makes the American premium exceed this European benchmark.